



Homework 11: We have so far considered the non-relativistic quantum field

$$\hat{\psi}(\mathbf{x}, t) = \int \frac{d^3 p}{(2\pi)^3} e^{i\mathbf{p} \cdot \mathbf{x}} \hat{a}_{\mathbf{p}}(t), \quad (1)$$

with

$$\hat{a}_{\mathbf{p}}(t) = e^{-i\epsilon_{\mathbf{p}} t} \hat{a}_{\mathbf{p}}(0), \quad \epsilon_{\mathbf{p}} = \frac{\mathbf{p}^2}{2m}. \quad (2)$$

In this homework, we ask what happens if we replace the non-relativistic dispersion relation by the relativistic one,

$$E_{\mathbf{p}} = \sqrt{\mathbf{p}^2 + m^2}. \quad (3)$$

We will see that this naturally leads to the Klein-Gordon equation, but also that a truly relativistic local quantum field cannot be built from annihilation operators alone.

Question 1: questions from the notes.

a) Show that the total particle number

$$\hat{N} = \sum_{\alpha} a_{\alpha}^{\dagger} a_{\alpha} \quad (4)$$

is conserved by any Hamiltonian made from terms with equal numbers of creation and annihilation operators, for example

$$\hat{H} = \sum_{\alpha, \beta} h_{\alpha\beta} a_{\alpha}^{\dagger} a_{\beta} + \frac{1}{2} \sum_{\alpha, \beta, \gamma, \delta} V_{\alpha\beta\gamma\delta} a_{\alpha}^{\dagger} a_{\beta}^{\dagger} a_{\gamma} a_{\delta} + \dots \quad (5)$$

Here

$$h_{\alpha\beta} = \langle \alpha | \hat{h} | \beta \rangle, \quad V_{\alpha\beta\gamma\delta} = \langle \alpha\beta | \hat{V} | \gamma\delta \rangle. \quad (6)$$

Start by proving that the commutators $[\hat{N}, a_{\alpha}^{\dagger}] = a_{\alpha}^{\dagger}$ and $[\hat{N}, a_{\alpha}] = -a_{\alpha}$.

b) Consider the free non-relativistic field Hamiltonian

$$\hat{H} = \int d^3 x \hat{\psi}^{\dagger}(\mathbf{x}) \left(-\frac{\nabla^2}{2m} + m \right) \hat{\psi}(\mathbf{x}). \quad (7)$$

The rest-energy term m has been included. Show that the single-particle energy $\hat{H} |\mathbf{p}\rangle = \epsilon_{\mathbf{p}} |\mathbf{p}\rangle$ is

$$\epsilon_{\mathbf{p}} = m + \frac{\mathbf{p}^2}{2m} \quad (8)$$

c) Rewriting the field in terms of the momentum-space annihilation operator show that

$$\hat{H} = \int \frac{d^3 p}{(2\pi)^3} \left(m + \frac{\mathbf{p}^2}{2m} \right) \hat{a}_{\mathbf{p}}^{\dagger} \hat{a}_{\mathbf{p}}. \quad (9)$$

Then show that the total momentum operator is conserved, $[\hat{\mathbf{P}}, \hat{H}] = 0$, where

$$\hat{\mathbf{P}} = \int \frac{d^3 p}{(2\pi)^3} \mathbf{p} \hat{a}_{\mathbf{p}}^{\dagger} \hat{a}_{\mathbf{p}} \quad (10)$$

You can start by showing that $[\hat{\mathbf{P}}, \hat{a}_{\mathbf{p}}^{\dagger}] = \mathbf{p} \hat{a}_{\mathbf{p}}^{\dagger}$, and $[\hat{\mathbf{P}}, \hat{a}_{\mathbf{p}}] = -\mathbf{p} \hat{a}_{\mathbf{p}}$.



Note: the total momentum operator $\hat{\mathbf{P}}$ is the generator of spatial translations, so it must be conserved by any translation-invariant Hamiltonian (as Eqs. (7) and (9) are). This is the *total* momentum of the system, as opposed to $\hat{\mathbf{p}}$, which is the momentum of a single particle/Fourier mode.

Question 2: A first attempt at a relativistic field.

- a) Let us attempt to define a relativistic field. We will call it a positive frequency field because it only contains annihilation operators, which have positive energy (we do not worry about positive vs negative frequency in the non-relativistic case, but it becomes an important distinction in relativistic theories). We define

$$\hat{\psi}_+(\mathbf{x}, t) = \int \frac{d^3p}{(2\pi)^3} e^{i\mathbf{p}\cdot\mathbf{x} - iE_{\mathbf{p}}t} \hat{a}_{\mathbf{p}}, \quad E_{\mathbf{p}} = \sqrt{\mathbf{p}^2 + m^2}. \quad (11)$$

Show that $\hat{\psi}_+$ satisfies the Klein-Gordon equation

$$(\partial_\mu \partial^\mu + m^2) \hat{\psi}_+ = \left(\frac{\partial^2}{\partial t^2} - \nabla^2 + m^2 \right) \hat{\psi}_+ = 0. \quad (12)$$

Note that the negative frequency field $\hat{\psi}_-$, which only contains creation operators,

$$\hat{\psi}_-(\mathbf{x}, t) = \int \frac{d^3p}{(2\pi)^3} e^{-i\mathbf{p}\cdot\mathbf{x} + iE_{\mathbf{p}}t} \hat{a}_{\mathbf{p}}^\dagger, \quad (13)$$

also satisfies the Klein-Gordon equation.

- b) Using $[\hat{a}_{\mathbf{p}}, \hat{a}_{\mathbf{p}'}^\dagger] = (2\pi)^3 \delta^3(\mathbf{p} - \mathbf{p}')$, show that the equal-time commutator is

$$[\hat{\psi}_+(\mathbf{x}, t), \hat{\psi}_+^\dagger(\mathbf{y}, t)] = \delta^3(\mathbf{x} - \mathbf{y}). \quad (14)$$

Now compute the commutator at two different times:

$$[\hat{\psi}_+(\mathbf{x}, t), \hat{\psi}_+^\dagger(\mathbf{y}, t')] = \int \frac{d^3p}{(2\pi)^3} e^{i\mathbf{p}\cdot(\mathbf{x}-\mathbf{y})} e^{-iE_{\mathbf{p}}(t-t')}. \quad (15)$$

Why is it a problem that the commutator does not vanish for spacelike separations, i.e. when $(t - t')^2 - (\mathbf{x} - \mathbf{y})^2 < 0$?



The condition that the commutator vanishes for spacelike separations is known as the microcausality condition, and is a fundamental requirement for any relativistic quantum field theory.

- c) Consider the Hermitian scalar field

$$\hat{\phi}(\mathbf{x}, t) = \int \frac{d^3p}{(2\pi)^3} \frac{1}{\sqrt{2E_{\mathbf{p}}}} \left(e^{i\mathbf{p}\cdot\mathbf{x} - iE_{\mathbf{p}}t} \hat{a}_{\mathbf{p}} + e^{-i\mathbf{p}\cdot\mathbf{x} + iE_{\mathbf{p}}t} \hat{a}_{\mathbf{p}}^\dagger \right). \quad (16)$$

Show that $\hat{\phi}^\dagger(\mathbf{x}, t) = \hat{\phi}(\mathbf{x}, t)$ and that $\hat{\phi}$ also satisfies the Klein-Gordon equation.

- d) Show that the Hermitian field $\hat{\phi}$ satisfies the microcausality condition, i.e. that $[\hat{\phi}(x), \hat{\phi}(y)] = 0$ for spacelike separations $(x - y)^2 < 0$.

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The field $\hat{\phi}$ is called a real scalar field, and is the simplest example of a relativistic quantum field (real because $\hat{\phi}^\dagger = \hat{\phi}$). The fact that it contains both creation and annihilation operators is crucial. One can also have a complex scalar field ($\hat{\Phi} \neq \hat{\Phi}^\dagger$), which contains two different creation and annihilation operators $\hat{a}_{\mathbf{p}}$ and $\hat{b}_{\mathbf{p}}$. It is used to describe a spin-0 particle and its corresponding antiparticle (note this implies we can distinguish them, for example, through their charge. Therefore, this is the appropriate field to use to describe charged spinless particles, such as a π^+ or π^-). In this case, microcausality requires that the field both destroy (create) particles (antiparticles):

$$\hat{\Phi}(\mathbf{x}, t) = \int \frac{d^3p}{(2\pi)^3} \frac{1}{\sqrt{2E_{\mathbf{p}}}} \left(e^{i\mathbf{p}\cdot\mathbf{x} - iE_{\mathbf{p}}t} \hat{a}_{\mathbf{p}} + e^{-i\mathbf{p}\cdot\mathbf{x} + iE_{\mathbf{p}}t} \hat{b}_{\mathbf{p}}^\dagger \right), \quad (17)$$

Microcausality is also intimately related to the discrete symmetries of charge conjugation, parity, and time reversal (CPT).